carbon.bsn

Purpose

carbon.bsn holds the basin-wide scalar coefficients for the SWAT+ carbon module. The reader fills org_frac, cb_wtr_coef, man_coef, and org_con (all defined in carbon_module.f90), the basin-scope tillage variables in tillage_data_module.f90 (bio_consf, till_consf, bmix_a/b/c, tillmix_a/b/c, till_eff_days), the photo_degrade_factor in plant_data_module.f90, and the five residue-decomposition tunables (n_act_frac, cnr_cap, cnr_ref, cpr_cap, cpr_ref) promoted from hard-coded constants into module-scope reals in carbon_module.f90.

carbon.bsn replaces the legacy carb_coefs.cbn (keyword-style) and works alongside the per-layer file carbon_lyr.bsn, which holds the rate constants and CO2 fractions that differ between the surface and subsurface layer groups.

Source

- Reader: src/carbon_bsn_read.f90
- Filename slot: src/input_file_module.f90, type input_basin, field carbon_bsn = "carbon.bsn".
- Target types: organic_fractions, carbon_water_coef, manure_coef, organic_controls in src/carbon_module.f90; module-scope reals n_act_frac, cnr_cap, cnr_ref, cpr_cap, cpr_ref in the same module; bio_consf, till_consf, till_eff_days, bmix_a/b/c, tillmix_a/b/c in src/tillage_data_module.f90; photo_degrade_factor in src/plant_data_module.f90.

Activation

The reader runs only when bsn_cc%cswat == 2 (set in codes.bsn). When cswat == 2 the file is **required**: if carbon.bsn (or its companion carbon_lyr.bsn) is missing or its data row cannot be parsed, the reader writes an error message to stdout and unit 9001 and calls error stop. There is no silent fallback to compiled defaults. carbon_bsn_read.f90 returns early for any other cswat value, so cswat = 1 (the reserved C-FARM slot) does not read this file.

The dynamic carbon model activates on cswat = 2, matching legacy SWAT numbering (1 = C-FARM, 2 = CENTURY). It was previously cswat = 1; existing codes.bsn files left at carbon = 1 now run with the carbon model off and must be changed to carbon = 2.

carbon.bsn is listed in the BASIN row of file.cio alongside codes.bsn and parameters.bsn (three columns). carbon_lyr.bsn rides at a fixed name beside it and does not appear in file.cio.

> **Note**
>
> The legacy cbn_diag first column is retired. Output gating now lives entirely in print.prt through the hru_cb_* and lsu_cb_* rows (see the carbon outputs reference).

Format

- Line 1: title (skipped).
- Line 2: column header (skipped).
- Line 3: 28 values, list-directed, read into the target fields in the order below. (The 28th value, mathers_method, was added when carbonDev merged to main; 27-value files from before the merge fail to parse and must add the trailing column.)

Values are space-delimited. Position is significant; the column header is decorative.

#	Field	Type	Target	Units	Description
1	init_seq	real	org_frac%frac_seq	fraction	initial fraction of total soil carbon held in the sequestered pool
2	init_microb	real	org_frac%frac_hum_microb	fraction	initial microbial-humus fraction of non-sequestered carbon
3	init_slow	real	org_frac%frac_hum_slow	fraction	initial slow-humus fraction of non-sequestered carbon
4	init_passive	real	org_frac%frac_hum_passive	fraction	initial passive-humus fraction of non-sequestered carbon
5	koc_c	real	cb_wtr_coef%prmt_21	none	partition coefficient for carbon loss in water and sediment (formerly prmt_21)
6	solc_ratio	real	cb_wtr_coef%prmt_44	0..1	ratio of soluble C in surface runoff to soluble C in percolate (formerly prmt_44)
7	till_eff_days	real	till_eff_days	days	number of days a tillage event remains effective
8	manure_c_frac	real	man_coef%rtof	none	manure organic N and P partition between fresh and stable pools (formerly rtof)
9	bio_consol	real	bio_consf	none	biological consolidation factor for soil moisture mixing
10	till_consol	real	till_consf	none	tillage consolidation factor for soil moisture mixing
11	tmpf_eqn	int	org_con%tmpf	none	temperature factor selector (1, 2, or 3) used in cbn_zhang2.f90
12	watf_eqn	int	org_con%watf	none	water factor selector (1 or 2) used in cbn_zhang2.f90
13	t_cbn_min	real	org_con%tn	deg C	minimum temperature bound for tmpf2 (lower limit of biological activity)
14	t_cbn_opt	real	org_con%top	deg C	optimum temperature for tmpf2
15	t_cbn_max	real	org_con%tx	deg C	maximum temperature bound for tmpf2 (upper limit of biological activity)
16	bmix_a	real	bmix_a	none	biological mixing curve coefficient A (mgt_tillfactor.f90 tillf option 4)
17	bmix_b	real	bmix_b	none	biological mixing curve coefficient B
18	bmix_c	real	bmix_c	none	biological mixing curve coefficient C
19	tillmix_a	real	tillmix_a	none	tillage mixing curve coefficient A
20	tillmix_b	real	tillmix_b	none	tillage mixing curve coefficient B
21	tillmix_c	real	tillmix_c	none	tillage mixing curve coefficient C
22	sfc_rsd_photodeg	real	photo_degrade_factor	1/day	surface residue photodegradation rate factor
23	n_act_frac	real	n_act_frac	fraction	fraction of organic N held in the active humus pool (drives active-to-stable flow in nut_nminrl.f90)
24	cnr_cap	real	cnr_cap	none	upper cap on residue C:N ratio before computing the decomposition factor
25	cnr_ref	real	cnr_ref	none	reference C:N ratio where the C:N decomposition factor equals 1
26	cpr_cap	real	cpr_cap	none	upper cap on residue C:P ratio before computing the decomposition factor
27	cpr_ref	real	cpr_ref	none	reference C:P ratio where the C:P decomposition factor equals 1

#	Field	Type	Target	Units	Description
28	`mathers_method`	int	`org_frac%mathers_method`	flag	slow-humus pool initialization: `0` original humus-slow init (default), `1` Mathers method (clay+silt based, from “Updating carbon pool initialization with DSSAT-CENTURY”). Used in `soil_nutcarb_init`.

The reader does not interpret column 1-4 as a probability simplex; the four `init_*` values are used as supplied. The four target fields hold the fractional partitioning of soil carbon into sequestered, microbial-humus, slow-humus, and passive-humus pools at simulation start.

The `tmpf_eqn` (col 11) and `watf_eqn` (col 12) integers select between the temperature and water response equations encoded in `cbn_zhang2.f90`. They are integers in the file but the reader does not enforce that; the consuming code treats them as `int(real)`.

Example

`refdata/Lafayette/carbon.bsn`:

```
carbon.bsn: basin-wide scalar carbon parameters (one record)
   init_seq   init_microb   init_slow   init_passive   koc_c   solc_ratio   till_eff_days   manure_c_frac
       0.95         0.02         0.44           0.54    1000          0.5           100            0.50         0.1
```

Calibration

Eighteen of the 28 fields are wired in `cal_parm_select.f90` and may be touched through `calibration.cal`: `init_seq`, `init_microb`, `init_slow`, `init_passive`, `koc_c`, `solc_ratio`, `manure_c_frac`, `bio_consol`, `till_consol`, `sfc_rsd_photodeg`, `n_act_frac`, `cnr_cap`, `cnr_ref`, `cpr_cap`, `cpr_ref`, `t_cbn_min`, `t_cbn_opt`, `t_cbn_max`. The six bmix/tillmix curve coefficients are also wired. The selector integers (`tmpf_eqn`, `watf_eqn`) and the integer-valued `till_eff_days` are not calibrated.

> **Note**
>
> `proc_cal` must run **before** `soil_nutcarb_init` for the `init_*` calibration values to take effect. The current `main.f90` already orders these correctly; do not reorder.

Related files

- `carbon_lyr.bsn`. Per-layer rate constants and CO2 allocation fractions.
- `codes.bsn`. The `cswat` switch that gates this file.
- `print.prt`. Output gating for the carbon families.
- Soil carbon (model theory). Equations and pool structure.
- Parameter change files. Calibration entries for the wired carbon parameters.